Denoising Trains the Potential Energy Surface

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Define the system. Data $x \in \mathbb{R}^d$ with density p. Throughout, ∇ denotes the gradient in $\tilde{x}$ and $\langle \cdot, \cdot \rangle$ the Euclidean inner product. Fix a noise scale (standard deviation) $\sigma > 0$. Draw unit noise $\varepsilon \sim \mathcal{N}(0, I)$, with I the $d \times d$ identity, independent of x, and perturb:

$$\tilde{x} = x + \sigma\varepsilon.$$

A network ε_θ with parameters θ sees $\tilde{x}$ and outputs a guess of the unit noise, trained on

$$\mathcal{L}_{\text{noise}}(\theta) \;=\; \mathbb{E}_{x,\varepsilon} \, \|\varepsilon_\theta(\tilde{x}) - \varepsilon\|^2 \,.$$

Let p_σ be the marginal density of $\tilde{x}$ and $q(\tilde{x} \,|\, x)$ the Gaussian perturbation kernel, which has variance σ^2 since $\tilde{x} - x = \sigma\varepsilon$:

$$p_\sigma(\tilde{x}) = \int p(x)\, q(\tilde{x} \,|\, x)\, dx, \qquad q(\tilde{x} \,|\, x) = (2\pi\sigma^2)^{-d/2} \exp\Big(-\tfrac{\|\tilde{x}-x\|^2}{2\sigma^2}\Big).$$

Consider p_σ as a Boltzmann density at inverse temperature $\beta = 1/(k_{\text{B}}T)$, where k_{B} is the Boltzmann constant and T the temperature:

$$p_\sigma(\tilde{x}) = \frac{e^{-\beta U_\sigma(\tilde{x})}}{Z_\sigma}, \qquad Z_\sigma = \int e^{-\beta U_\sigma(\tilde{x})}\, d\tilde{x},$$

so the true (noised) potential is $U_\sigma = -\frac{1}{\beta}\log p_\sigma - \frac{1}{\beta}\log Z_\sigma$. Let the model potential be U_θ with density $q_\theta = e^{-\beta U_\theta}/Z_\theta$, $Z_\theta = \int e^{-\beta U_\theta}\, d\tilde{x}$, and induced noise predictor $\varepsilon_\theta = \sigma\beta\, \nabla U_\theta$. Define the potential-matching (force-mismatch) loss

$$\mathcal{L}_{\text{PES}}(\theta) \;\coloneqq\; \mathbb{E}_{\tilde{x}\sim p_\sigma} \, \|\nabla U_\theta(\tilde{x}) - \nabla U_\sigma(\tilde{x})\|^2 \,.$$

Theorem. *For every $\sigma > 0$, the noise loss is a positive affine image of this loss, so the two losses are minimised by exactly the same parameters:*

$$\boxed{\arg\min_\theta \, \mathcal{L}_{\text{noise}}(\theta) \;=\; \arg\min_\theta \, \mathcal{L}_{\text{PES}}(\theta)}$$

Training to predict noise is training the potential energy surface to be close to ideal up to an additive constant.

Step 1 – the minimiser is the conditional mean

Proof. Let $m(\tilde{x}) \coloneqq \mathbb{E}[\varepsilon \,|\, \tilde{x}]$ be the conditional mean of the unit noise given the perturbed point. Split each error around it:

$$\begin{aligned} \|\varepsilon_\theta(\tilde{x}) - \varepsilon\|^2 &= \big\|\big(\varepsilon_\theta(\tilde{x}) - m(\tilde{x})\big) + \big(m(\tilde{x}) - \varepsilon\big)\big\|^2 \\ &= \underbrace{\|\varepsilon_\theta(\tilde{x}) - m(\tilde{x})\|^2}_{A} + 2\,\langle \varepsilon_\theta(\tilde{x}) - m(\tilde{x}),\, m(\tilde{x}) - \varepsilon\rangle + \underbrace{\|m(\tilde{x}) - \varepsilon\|^2}_{C}, \end{aligned}$$

where A and C name the two squared terms. Take $\mathbb{E}[\,\cdot \mid \tilde{x}]$. Given $\tilde{x}$, the vector $\varepsilon_\theta(\tilde{x}) - m(\tilde{x})$ is fixed, and

$$\mathbb{E}\left[\, m(\tilde{x}) - \varepsilon \,\middle|\, \tilde{x} \,\right] = m(\tilde{x}) - \mathbb{E}[\varepsilon \mid \tilde{x}] = 0, \qquad \text{[def. of } m\text{]}$$

so the cross term vanishes. Taking the outer expectation $\mathbb{E}_{\tilde{x}}$ of what remains,

$$\mathcal{L}_{\text{noise}}(\theta) \;=\; \mathbb{E}_{\tilde{x}} \left\|\varepsilon_\theta(\tilde{x}) - m(\tilde{x})\right\|^2 \;+\; \mathbb{E}\left\|m(\tilde{x}) - \varepsilon\right\|^2. \qquad \text{[law of iterated expectation]}$$

The second term equals the constant $\kappa \coloneqq \mathbb{E}\left\|m(\tilde{x}) - \varepsilon\right\|^2$ (the irreducible noise floor, independent of θ); the first is ≥ 0. Hence

$$\arg\min_\theta \mathcal{L}_{\text{noise}}(\theta) = \arg\min_\theta \mathbb{E}_{\tilde{x}} \left\|\varepsilon_\theta(\tilde{x}) - m(\tilde{x})\right\|^2,$$

and the minimiser drives the first term to 0, giving

$$\boxed{\varepsilon_\theta^\star(\tilde{x}) = m(\tilde{x}) = \mathbb{E}[\varepsilon \mid \tilde{x}] \quad \text{for } p_\sigma\text{-a.e. } \tilde{x}.}$$

$\square$

Step 2 – conditional mean = scaled score

For the Gaussian perturbation kernel, using $\tilde{x} - x = \sigma\varepsilon$,

$$\nabla \log q(\tilde{x} \mid x) = -\frac{\tilde{x} - x}{\sigma^2} = -\frac{\varepsilon}{\sigma} \quad \Longrightarrow \quad \nabla q(\tilde{x} \mid x) = -\frac{\tilde{x} - x}{\sigma^2}\, q(\tilde{x} \mid x).$$

Lemma 1. $\nabla \log p_\sigma(\tilde{x}) = -\dfrac{1}{\sigma}\, \mathbb{E}[\varepsilon \mid \tilde{x}]$.

Proof. $q(\cdot \mid x)$ and $\nabla q(\cdot \mid x)$ are bounded above by an integrable function uniformly on compacta, so we may differentiate under the integral:

$$\begin{aligned}
\nabla \log p_\sigma(\tilde{x}) &= \frac{\nabla p_\sigma(\tilde{x})}{p_\sigma(\tilde{x})} && \text{[chain rule]} \\
&= \frac{1}{p_\sigma(\tilde{x})} \int p(x)\, \nabla q(\tilde{x} \mid x)\, dx && \text{[Leibniz]} \\
&= \int \frac{p(x)\, q(\tilde{x} \mid x)}{p_\sigma(\tilde{x})} \left(-\frac{\tilde{x} - x}{\sigma^2}\right) dx && \text{[Gaussian grad.]} \\
&= -\frac{1}{\sigma^2}\, \mathbb{E}\left[\, \tilde{x} - x \,\middle|\, \tilde{x} \,\right] && \text{[Bayes: integrand is the posterior } p(x \mid \tilde{x})\text{]} \\
&= -\frac{1}{\sigma^2}\, \mathbb{E}[\sigma\varepsilon \mid \tilde{x}] = -\frac{1}{\sigma}\, \mathbb{E}[\varepsilon \mid \tilde{x}]. && [\tilde{x} - x = \sigma\varepsilon]
\end{aligned}$$

$\square$

Equivalently $m(\tilde{x}) = \mathbb{E}[\varepsilon \mid \tilde{x}] = -\sigma\, \nabla \log p_\sigma(\tilde{x})$.

Step 3 – affine equivalence of the two losses

Proof. Define the implied score network $s_\theta(\tilde{x}) := -\sigma^{-1}\varepsilon_\theta(\tilde{x})$, i.e. $\varepsilon_\theta = -\sigma s_\theta$. Using $\varepsilon = (\tilde{x}-x)/\sigma = -\sigma\nabla \log q(\tilde{x}\,|\,x)$,

$$\mathcal{L}_{\text{noise}}(\theta) = \mathbb{E}\,\|-\sigma s_\theta(\tilde{x}) + \sigma\nabla \log q(\tilde{x}\,|\,x)\|^2 = \sigma^2\ \mathbb{E}\,\|\,s_\theta(\tilde{x}) - \nabla \log q(\tilde{x}\,|\,x)\,\|^2\,.$$

Decompose around $\nabla \log p_\sigma(\tilde{x})$. The cross term has conditional mean $\mathbb{E}[\nabla \log q(\tilde{x}\,|\,x)\,|\,\tilde{x}] = \nabla \log p_\sigma(\tilde{x})$ by the Lemma, hence vanishes, leaving

$$\mathcal{L}_{\text{noise}}(\theta) = \sigma^2\Big(\underbrace{\mathbb{E}\,\|\,s_\theta(\tilde{x}) - \nabla \log p_\sigma(\tilde{x})\,\|^2}_{\mathcal{L}_{\text{score}}(\theta)} + \underbrace{\mathbb{E}\,\|\nabla \log p_\sigma(\tilde{x}) - \nabla \log q(\tilde{x}\,|\,x)\|^2}_{C'}\Big).$$

C' is constant in θ, so $\mathcal{L}_{\text{noise}}$ is an affine image of the score-matching loss $\mathcal{L}_{\text{score}}$ with positive slope σ^2, i.e. $t \mapsto \sigma^2 t + \sigma^2 C'$ for the dummy argument t. A positive affine map preserves minimisers:

$$\arg\min_\theta\ \mathcal{L}_{\text{noise}}(\theta) = \arg\min_\theta\ \mathcal{L}_{\text{score}}(\theta).$$

At the optimum $s_\theta^\star = \nabla \log p_\sigma$, equivalently $\varepsilon_\theta^\star = -\sigma\,\nabla \log p_\sigma$.

Now put in the Boltzmann constants from the main theorem. With $\beta = 1/(k_{\text{B}}T)$ and $p_\sigma = e^{-\beta U_\sigma}/Z_\sigma$, $Z_\sigma = \int e^{-\beta U_\sigma}\,d\tilde{x}$,

$$\log p_\sigma = -\beta U_\sigma - \log Z_\sigma \quad\Longrightarrow\quad \nabla \log p_\sigma = -\beta\,\nabla U_\sigma \qquad [\nabla \log Z_\sigma = 0,\ \text{constant in } \tilde{x}]$$

i.e. the score is the force divided by $k_{\text{B}}T$: $\nabla \log p_\sigma = -\frac{1}{k_{\text{B}}T}\nabla U_\sigma$. For the model $q_\theta = e^{-\beta U_\theta}/Z_\theta$, $Z_\theta = \int e^{-\beta U_\theta}\,d\tilde{x}$, likewise

$$s_\theta = \nabla \log q_\theta = -\beta\,\nabla U_\theta, \qquad \varepsilon_\theta = -\sigma s_\theta = \sigma\beta\,\nabla U_\theta = \frac{\sigma}{k_{\text{B}}T}\,\nabla U_\theta,$$

matching the main theorem. Substituting $s_\theta = -\beta\nabla U_\theta$ and $\nabla \log p_\sigma = -\beta\nabla U_\sigma$ into the defined score loss,

$$\mathcal{L}_{\text{score}}(\theta) = \mathbb{E}\,\|-\beta\nabla U_\theta + \beta\nabla U_\sigma\|^2 = \beta^2\,\mathcal{L}_{\text{PES}}(\theta) = \frac{1}{(k_{\text{B}}T)^2}\,\mathcal{L}_{\text{PES}}(\theta),$$

so the aforementioned identity becomes the main theorem's identity,

$$\mathcal{L}_{\text{noise}}(\theta) = \sigma^2\big(\beta^2\mathcal{L}_{\text{PES}}(\theta) + C'\big) = \frac{\sigma^2}{(k_{\text{B}}T)^2}\,\mathcal{L}_{\text{PES}}(\theta) + \sigma^2 C', \qquad \mathcal{L}_{\text{PES}}(\theta) \geq 0.$$

Since $\sigma^2/(k_{\text{B}}T)^2 > 0$ and $\sigma^2 C'$ is constant in θ, the two losses have identical minimisers, and

$$\boxed{\arg\min_\theta\ \mathcal{L}_{\text{noise}}(\theta) = \arg\min_\theta\ \mathcal{L}_{\text{PES}}(\theta).}$$

$\square$

thus,

$$\mathcal{L}_{\text{PES}}(\theta) = 0 \iff \nabla U_\theta = \nabla U_\sigma\ \ p_\sigma\text{-a.e.} \iff U_\theta = U_\sigma + c\ \ (\text{some constant } c) \iff q_\theta = p_\sigma.$$

The energy-zero constant c is *not* pinned: if $U_\theta = U_\sigma + c$ then $Z_\theta = \int e^{-\beta(U_\sigma+c)} = e^{-\beta c} Z_\sigma$, so

$$q_\theta = \frac{e^{-\beta(U_\sigma+c)}}{Z_\theta} = \frac{e^{-\beta c}\, e^{-\beta U_\sigma}}{e^{-\beta c} Z_\sigma} = \frac{e^{-\beta U_\sigma}}{Z_\sigma} = p_\sigma$$

for every c, meaning that the partition function absorbs the shift. So the densities and forces are recovered, while the absolute energy is fixed only up to c (equivalently, up to the free-energy offset $k_\mathrm{B} T \log Z$). Therefore the minimiser $\theta^\star := \arg\min_\theta \mathcal{L}_\mathrm{noise}(\theta)$ is, *simultaneously*,

(i) the point where the learned noise best matches the true noise added;

(ii) the point where the learned PES is closest to the true PES: $\mathcal{L}_\mathrm{PES}$ is minimised, and (at full capacity) the surfaces coincide up to the constant c.

Scaling Factors

Single noise level: The factor is $\sigma^2/(k_\mathrm{B}T)^2 > 0$, a positive constant. It cannot move $\arg\min_\theta$, only rescales the loss, and hence the gradient magnitude (an effective learning-rate change). *At a fixed σ and T, adding or removing a scalar weight does nothing to the optimum.*

Many noise levels: In practice one network is shared across a range of scales. Each minibatch draws a noise level from a *sampling distribution* π over σ — i.e. π is the rule that decides how often each σ is trained on (e.g. uniform in $\log \sigma$, or a fixed schedule). One may also apply an explicit weight $\gamma(\sigma) \geq 0$ to each level. The total training objective is then

$$\mathcal{J}(\theta) = \mathbb{E}_{\sigma\sim\pi}\left[\gamma(\sigma)\, \mathcal{L}^{(\sigma)}_\mathrm{noise}(\theta)\right] = \mathbb{E}_{\sigma\sim\pi}\Bigg[\underbrace{\frac{\gamma(\sigma)\,\sigma^2}{(k_\mathrm{B}T)^2}}_{\lambda(\sigma)}\; \mathcal{L}^{(\sigma)}_\mathrm{PES}(\theta)\Bigg] \;+\; \underbrace{\mathbb{E}_{\sigma\sim\pi}\big[\gamma(\sigma)\sigma^2 C'_\sigma\big]}_{\text{constant in }\theta},$$

where $\mathcal{L}^{(\sigma)}_\mathrm{noise}(\theta) = \mathbb{E}\,\|\varepsilon_\theta - \varepsilon\|^2$ is the noise loss at level σ and $\mathcal{L}^{(\sigma)}_\mathrm{PES}(\theta) = \mathbb{E}_{p_\sigma}\,\|\nabla U_\theta - \nabla U_\sigma\|^2$ the matching force loss. Here the question is *which noise levels the network pays attention to*: each level pulls on the shared potential with effective weight

$$\boxed{\lambda(\sigma) \;=\; \frac{\gamma(\sigma)\,\sigma^2}{(k_\mathrm{B}T)^2}.}$$

With fixed T, the factor $1/(k_\mathrm{B}T)^2$ is a global constant and drops out of the *relative* weighting, leaving $\lambda(\sigma) \propto \gamma(\sigma)\,\sigma^2$. So even with no explicit weight ($\gamma \equiv 1$), the plain noise loss already imposes $\lambda \propto \sigma^2$. Large σ means heavy blurring (coarse/global), while small σ begets fine detail. Thus out of the box the network spends most of its effort on coarse structure and comparatively neglects near-data detail. *Choosing a custom $\gamma(\sigma)$ is therefore correcting the scaling that is already baked in.*

- Equal attention to every scale: $\gamma(\sigma) = (k_\mathrm{B}T)^2/\sigma^2$, cancelling the σ^2 so $\lambda \equiv 1$.
- Sharper samples / accurate forces near the data: up-weight small σ, then taper off eventually with something like $\lambda(\sigma) = 1 + \frac{a}{1+(\sigma/\sigma_0)^2}$ where $1 + a$ for $a > 0$ is the height of the scaling and $\sigma_0 > 0$ controls the taper-off rate.

- Potentially adding σ as a parameter of $T := T(\sigma)$ could assist in relative weighting through simultaneous annealing with noise.

As $\sigma \to 0$, $p_\sigma \to p$ and $\nabla \log p_\sigma \to \nabla \log p$, but the implied force target $\nabla U_\theta = (k_\mathrm{B}T/\sigma)\,\varepsilon_\theta$ has target $\sim (k_\mathrm{B}T/\sigma)\,\varepsilon$, whose variance is $\sim (k_\mathrm{B}T)^2/\sigma^2$, so regressing the force (or score) directly is ill-conditioned at small noise. Predicting the unit noise ε keeps the target $O(1)$ at every scale and pulls σ and $k_\mathrm{B}T$ out into the analytic factor $\sigma^2/(k_\mathrm{B}T)^2$. ε-prediction has *already* chosen $\lambda(\sigma) \propto \sigma^2$ for you, so any explicit $\gamma(\sigma)$ multiplies on top of that.